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SCSE 2123 SOFTWARE REQUIREMENTS ENGINEERING

TOPIC 6: Practices for Model-Based Requirement Documentation (Part 2)

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Contents

- Introduction to Model-Based Work Products
- Modeling Structure and Data – Domain Model and Class Diagram
- Modeling State and Behavioral –State Diagram

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Objectives

At the end of this topic, students will be able to:

1.Design and construct domain class diagram, class diagram and state diagram ensuring consistency with user requirements.

2. Evaluate the completeness and correctness of model-based work products in representing system requirements using in-class exercises and peer review

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6.1 REQUIREMENTS MODELLING PERSPECTIVES

RECAP

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Aspects of Functional Requirements

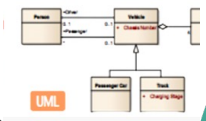
RECAP

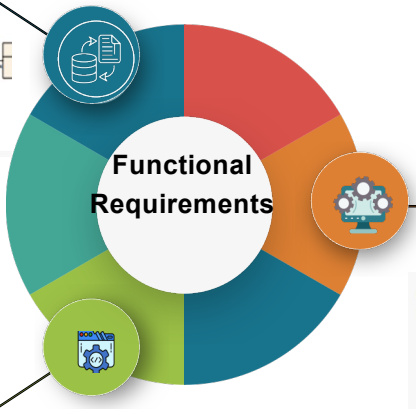
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Structure and Data

- Static structure of a system and the (persistent) data
- What a system must know in order to perform the required functions and deliver the required results.
- Includes static relationships of data.

6.1(d)



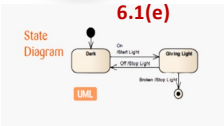


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State and Behavioural

- State dependent behaviour of a system.
- How a system shall react to which external event.

6.1(e)

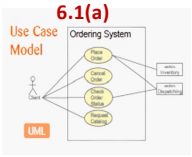


3

Functions and Flow

- Functions that a system shall provide.
- Flow of control and data within and between functions.
- What systems need for creating the required results from given inputs.

6.1(a)



6.1(c)



6.1(b)



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6.1 (d) MODELLING STRUCTURE AND DATA

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Modelling Structure and Data

Structure and Data Model

- Models that focus on structure and data aspects specify requirements for static structural properties of
  - A system or
  - A domain

Definition

Domain Model

- Model describing phenomena in application domain.
- A Domain Class Diagram describes the types of objects in the problem domain and their relationships.
- Unlike diagrams used in the design phase, it captures the *"vocabulary"* of the business, not the software implementation.

Class Model

- A model consisting of a set of classes and relationships between them.

Key Purpose

- "To verify that we understand the business concepts and how they link together before we write a single line of code.

Common Models for Structure and Data

Static Domain Models represented by

- UML class diagram
- Entity relationship diagrams (ERD)

Class Models represented by

- UML class diagram
- Entity relationship diagrams (ERD)

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- Block Definition Diagram

UML Class Diagrams

- UML class diagrams are a common approach to modeling the data perspectives of requirements
- A class diagram consist of a set of classes and the associations between these classes.
- As a conceptional model it is also called domain class model or domain model

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Domain Model vs. Design Model

Domain Model (Requirements)

- Focus: Real-world concepts
- Attributes: Business data
- Methods: None
- Visibility: Not relevant

Design Model (Implementation)

- Focus: Software components
- Attributes: Data type (int, string etc)
- Methods: Getter, Setter, logic
- Visibility: Public, Private

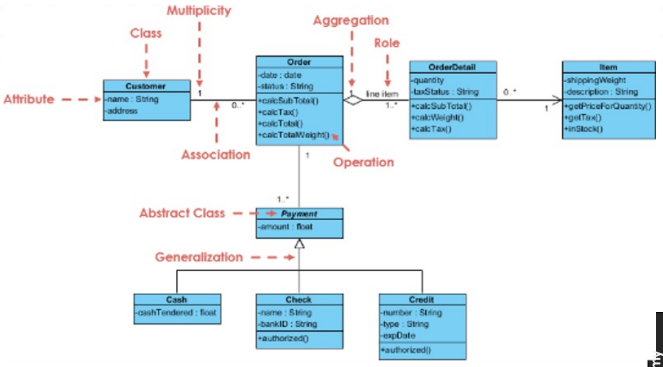
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The Notation: Anatomy of a Class

At the domain level, a class is represented by a simple rectangle with two main compartments

- Class Name: A noun, starting with a capital letter (e.g., Customer, Order).
- **Attributes:** The detailed information we need to know about the entity (e.g., *emailAddress*, *dateOfBirth*).
- **No Methods:** We intentionally leave out operations because we are modeling *data* structure, not behavior.



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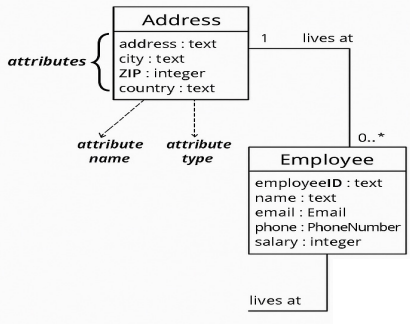
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Example UML Class Diagram

[class name]
[attribute name]
....

A class describes a set of (material or immaterial) objects that have a similar structure, behaviour and relationships.

An attribute describes a property of the class. The attribute is expressed as a certain type that restricts e values that can be assigned to the attribute.



Example UML Class Diagram

[class name]

A class describes a set of (material or immaterial) objects that have a similar structure, behaviour and rrelaton-ships.

[name]

An association relates two classes to each other.

Multiplicities

Define how many instances of a class at the corresponding association end can can participate in the association to a given instance of the class at the other end of the association:
0.. 1 (zero or one time)
0.. * (zero or more times)
1 .. * (one or more times)
7 (exactly seven times)
1 (default value, exactly one time)



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Domain classes and attributes

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Domain Class:

- Describes an object, a concept, a place or a person from the domain

Attribute:

- Domain (Data) item that is contained equally in every object of a class and is represented by every object with an individual value

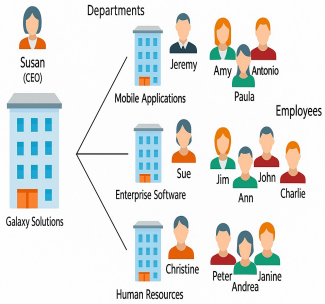


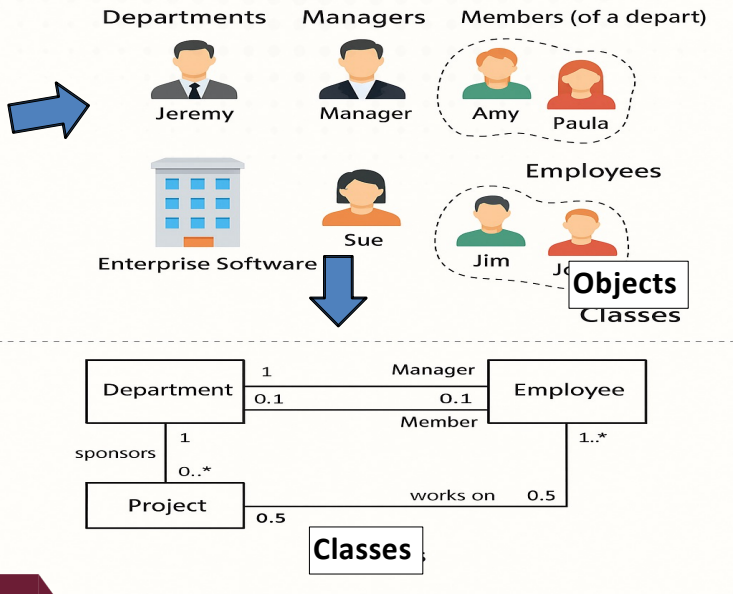
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Example UML Class Diagram

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Example Domain: Software Company

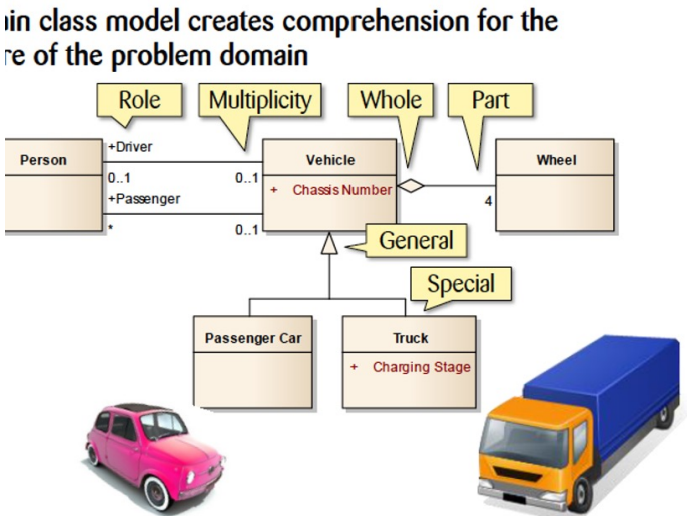


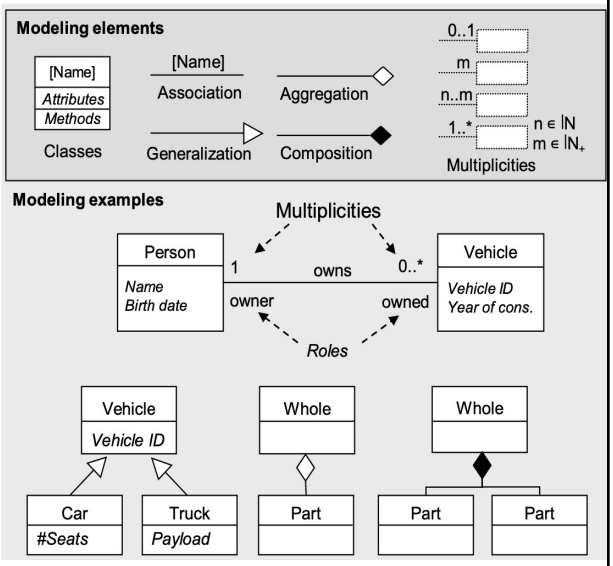


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Example of UML Class Diagram





UML Domain class diagram – elements (ebook, page 74)

K. Pohl and C. Rupp, Requirements Engineering Fundamentals, 1st edition. CA, USA: Rocky Nook, 2011.

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Relationships

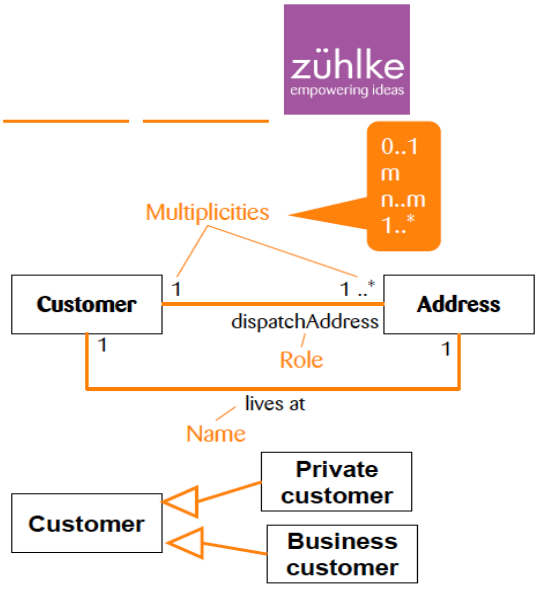
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Association

- represents a relationship between **objects** of one or several classes.
- At the ends of an association path multiplicities can indicate how many instances of the classes can be related to each other

Generalization

- generalization relation between classes ("is a" relation)



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Relationships

Aggregation:

- Whole-part relation
- Part can exist without whole



Composition

- Whole-part relation
- Part cannot exist without whole



- More video example "MOOC UML #11: Creating a Class Diagram: Example University Information System" at https://www.youtube.com/watch?v=tKy85WODAUk

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Example: Library System

A transcript from a meeting with the Head of Librarian

"We need a system to track our collection. Our library has thousands of books. For each book, we need to know the ISBN, title, and author.
However, we often buy multiple copies of popular books. Each physical copy has a unique barcode so we can tell them apart.
Members sign up with their name and email. A member can borrow physical copies. When they do, we create a loan record. The loan must track the checkout date and the due date.
If a copy is not returned on time, a fine is generated for that member. The fine has an amount and a status (paid or unpaid)."

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Steps to design domain class diagram

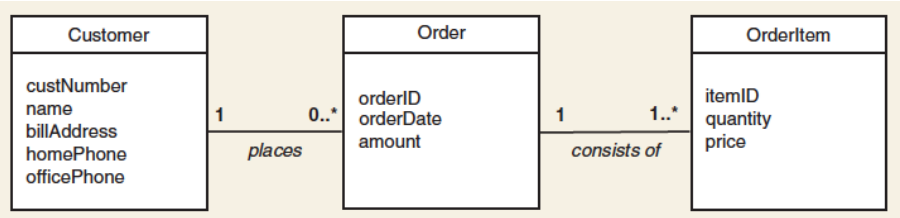
1.Identify Noun
2.Assign Attributes to the identified noun
3.Identify relatiionship between class

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Activity 6.1(d)-1

Interpret the following domain model class diagram:

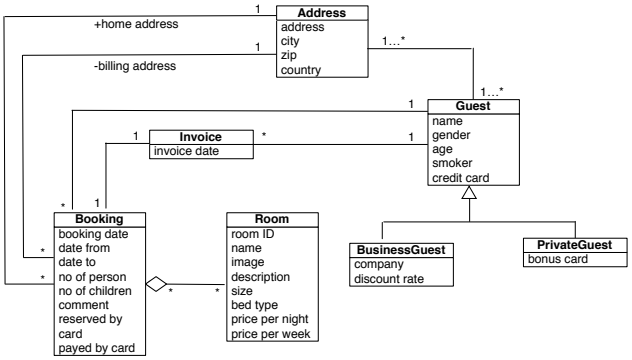


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Activity 6.1(d)-2

- The following Figure models the data structures and its relationships for the proposed new hotel booking system. Based on the model, identify and explain TWO (2) different types of relationships between classes (***generalization, association, and aggregation***)



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Further use of the domain class model

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In analysis and design

- Deriving software class models
- Deriving the database schema/model
- Identification of interfaces between systems
  - synchronization of classes, used in two or more systems

In implementation and test

- Deriving test cases
  (e.g. valid relations, value ranges)
- Deriving data access authorization

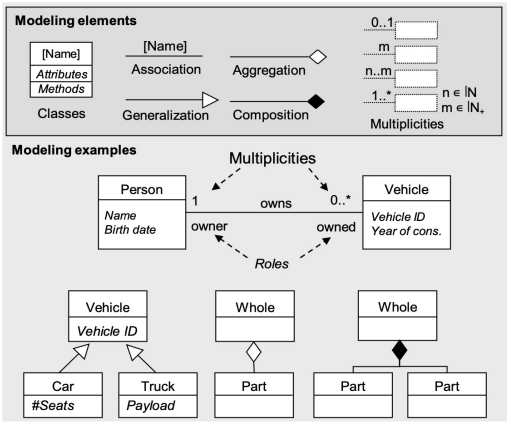
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Topic 6(d) – Summary

1.Data perspective – Class diagram
2.In this course, we focus on domain class diagram use to model the structure of a system - class is an entity representation in the real-world
3.Domain class diagram elements – classes (with attributes only), relationships and multiplicities



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6.1 (e) MODELLING STATE & BEHAVIOR

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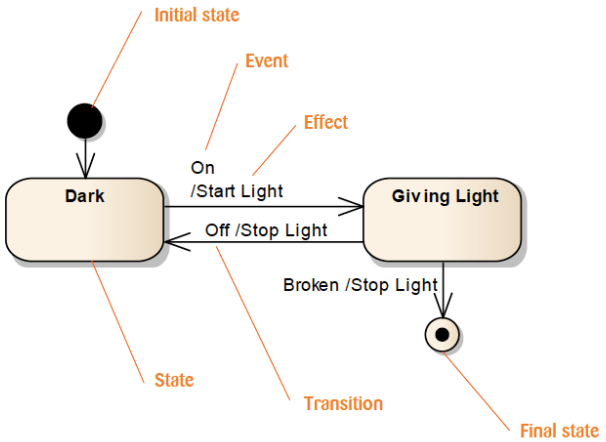
State Transition Diagram

- Models that focus on state and behavior. It specify requirements for the behavior of a system or a domain component in terms of
  - State dependent reactions to events
  - Dynamics of component interaction
- **state changes** of a **single object** according to **events** and **messages** which the **class sends and receives.**
- **A State Transition Diagram (or State Machine Diagram) is a behavioral model that depicts the discrete states an object undergoes during its lifetime. It maps out how an object reacts to specific events based on its current state.**
- Rules of thumb:
  - Not all classes need this. It is essential for complex, reactive objects—like Orders, Invoices, or Bookings—where the valid operations depend heavily on the object's current status (e.g., you can't "ship" an order that hasn't been "paid").

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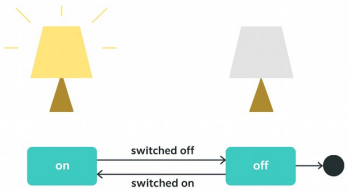
Example 1 – State diagram





1.Observe the symbols/notations used
2.Guess what object that this example 1 represent?

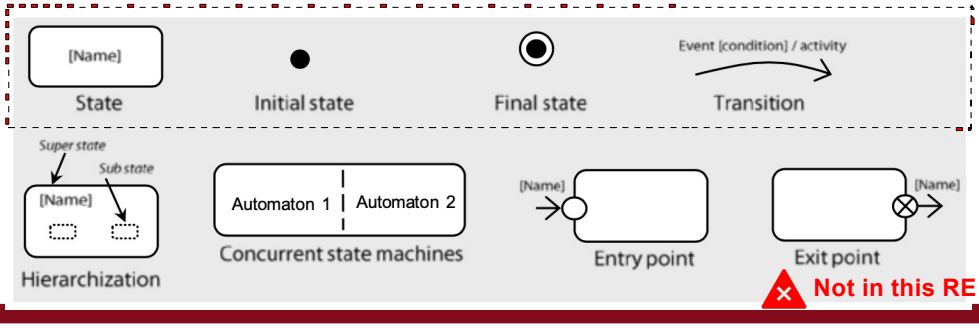
Simple State Diagram - Example



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Simple Notation



- **Initial State: Represented by a solid filled black circle. Indicates the starting point of the object's lifecycle.**
- **State: Represented by a rectangle with rounded corners. The name of the state is written inside**
- **Final State: A circle surrounding a smaller solid circle (bullseye). Indicates the end of the lifecycle.**
- **Transition: A solid line with an arrowhead connecting states, indicating the direction of flow.**

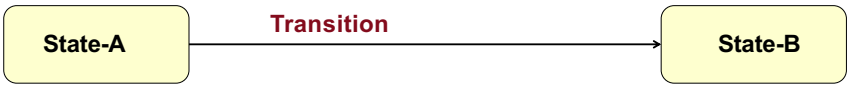
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Elements of State Diagram

1.State:
- a condition during the life of an object when it satisfies some condition, performs some action, or waits for an event
- found by examining the attributes and links defined for the object
- represented as a rectangle with rounded corners

2.Transition
- represents a change of the internal condition/state of an object



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Transition notation



i.Event:

- triggers the transition or change (e.g. Order Placed)

ii.Condition/Guard:

- **A boolean condition that must be true for the transition to occur. (e.g., *"[In Stock]"*)**
- Transition only eligible to fire when guard evaluates to true
- Guards of transition exiting one state are mutually exclusive

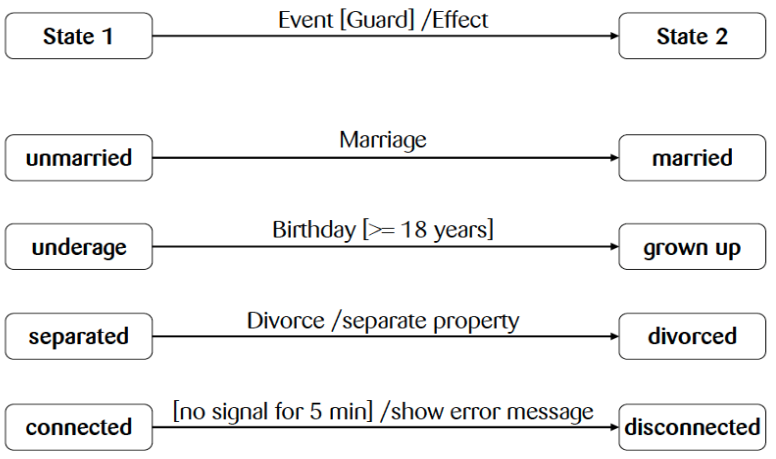
iii.Action/Effect:

- **An activity executed during the transition. (e.g., "/ Update Inventory")**
- May include operation calls, the creation or destruction of another objects, or the sending of a signal to an object.

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Examples of State Transitions



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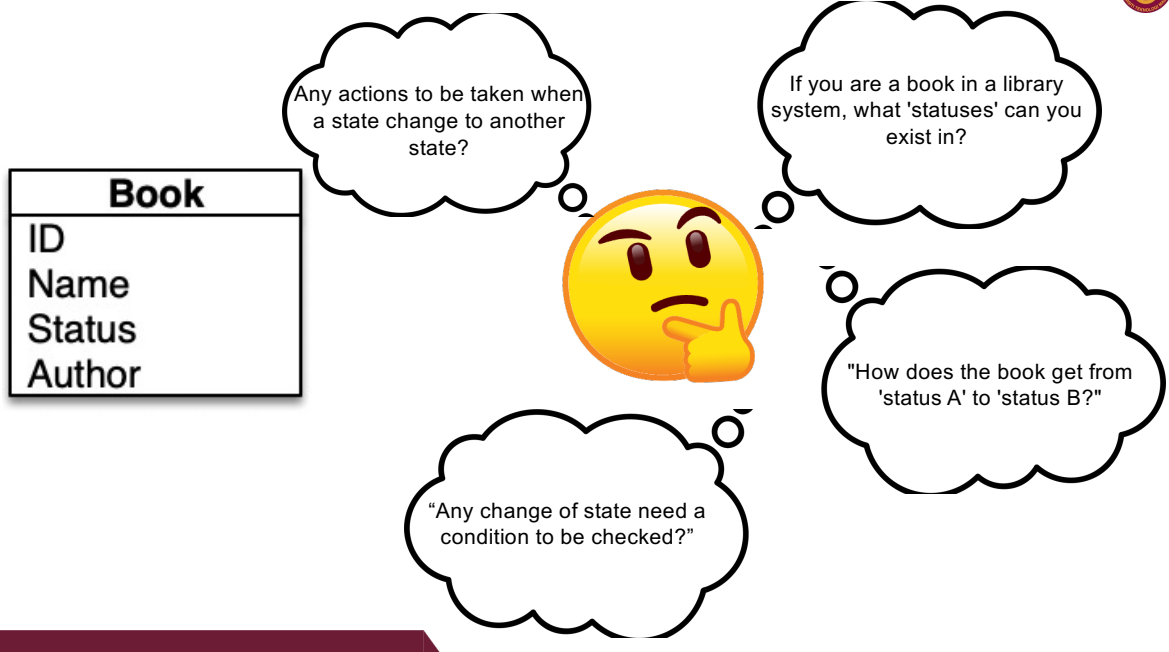
Steps to design state transition diagram

1. Identify state for the object. Tips: can think of the "status" of the object.
2. Identify what makes the state change to another state
3. Identify guards (conditions) that prevent transitions - for a possible transition

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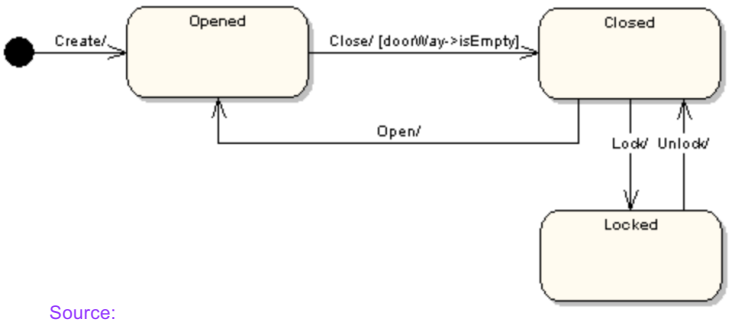
Example



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Example 2 – State diagram for train (LRT) door



Source:
https://sparxsystems.com/resources/tutorials/uml2/state-diagram.html



Just-in-time information @ enough to know 2 elements:

1.state
2.transition

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More on Events types

i. Call events:
- The dispatch of an operation.
- The control passes from the sender to the receiver object.
- The transition is triggered by the event.
- The operation is completed, the receiver transitions to a new state, and control returns to the sender.

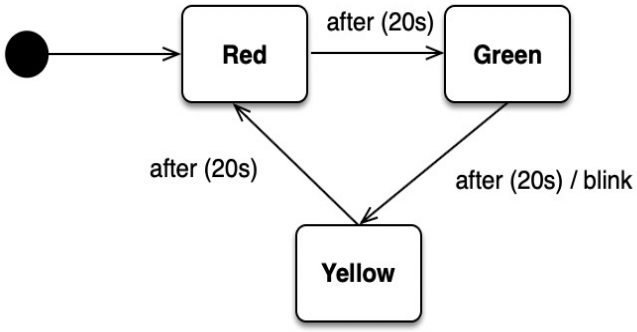
ii. Time events:
- The passage of time.
- In UML, represented by the keyword eg. After followed by some expression that evaluates to a period of time; after (2 seconds)

iii. Change events:
- Change in value of some entity or a condition becoming true.
- Is shown as a guard-condition on a state transition.
- For example: false(hasProblems) or true(hasProblems)

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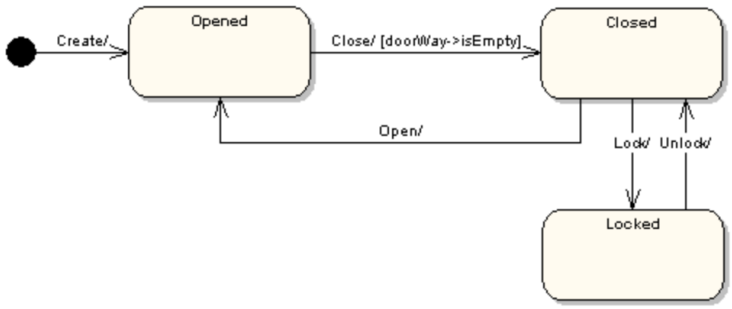
Example time event – State transition diagram for traffic light



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Recap on Example 2 – State diagram for train (LRT) door



1.Call events ~ operation by an object
2.Time event ~ something happen upon time
3.Change event ~ something happen that change object's value/condition

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Common problem

Can I draw a state diagram for the whole system?

- **No** - a state diagram normally only models the behavior of the objects of a single class. One of the most common mistakes that students make when learning the technique is to try to model the behavior of the whole system in a single state diagram.
- State transition diagrams are created when:
  - A class has a complex life cycle.
  - An instance of a class may update its attributes in a number of ways through the life cycle.
  - A class has an operational life cycle.
  - Two classes depend on each other.

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In- Class Activity 1

Draw the state transition for a luggage belt class. The belt is started when the start button is pressed and runs until either the stop button is pressed or until there is no luggage on the belt. This condition is when no luggage has been detected in the previous 60 seconds.

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In- Class Activity 2

- Draw the state transition diagram for a simple Battery Charger that can charge two batteries in parallel. The charger has three modes: `idle`, `discharging`, and `charging`.
- A button can be pressed (`evStart`) to start charging the batteries. However, before each battery can be charged, it must be discharged.
- When each battery is discharged, it sends an event (`evBatteryA_Discharged` or `evBatteryB_Discharged`) to the Battery Charger.
- When each battery is charged, it sends an event (`evBatteryA_Full` or `evBatteryB_Full`) to the Battery Charger.
- When both batteries are charged, the Battery Charger returns to the idle mode.

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